

Convention: power rails above components; GND below.

The schematic diagram illustrates the display side of the Volthium reader. It features an ESP32-S3-WROOM-1-N16R8 microcontroller at the center. Key components and connections include:

- Power and Ground:** Power rails (VCC, 3V3, GND) are shown at the top. Ground connections are at the bottom.
- USB-C Connector (J1):** Connected to the microcontroller's USB pins (USB_D+, USB_D-, USB_DP, USB_DM).
- USB-to-UART Bridge (SN65HVD3082E):** Connected to the microcontroller's UART pins (UART_TX, UART_RX, DE, RE, RD).
- 3.3V Voltage Divider:** A network of resistors (R1, R2, R3, R4) and capacitors (C1, C2, C3, C4, C5, C6, C7, C8, C9, C10) used to provide a stable 3.3V supply to the microcontroller.
- Push Buttons (BTN1, BTN2, BTN3):** Connected to the microcontroller's GPIO pins (BTN1_IN, BTN2_IN, BTN3_IN).
- Other Components:** Various other components like capacitors (C1-C10), resistors (R1-R5), and diodes (V12-CATSE, V12-PROT) are shown.

The schematic is labeled 'display_side.kicad_sch' and is part of a project titled 'Volthium reader - display side'.